Table 1: Generating factors of the model zoos. Several values for each parameter define the grid. `Arch` denotes the architecture: CNN (s) - small CNN architecture, CNN (m) - medium CNN architecture, RN-18 - ResNet-18. `Init` denotes the initalization methods: U - uniform, N - normal, KU - Kaiming Uniform, KN - Kaiming Normal. `Activation` denotes the activation function: T - Tanh, S - Sigmoid, R - ReLU, G - GeLU. `Optim` denotes the optimizer: AD - Adam, SGD - Stochastic Gradient Descent. Models with learning rates denoted with * have been trained with a one-cycle LR scheduler, the listed LR is the maximum value.

Dataset	Arch	Config	`Init`	`Activation`	`Otpim`	`LR`	`WD`	`Dropout`	`Seed`
MNIST	CNN (s)	Seed	U	T	AD	3e-4	0	0	1-1000
	CNN (s)	Hyp-10-r	U, N, KU, KN	T, S, R, G	AD, SGD	1e-3,1e-4	1e-3, 1e-4	0, 0.5	~ 10
	CNN (s)	Hyp-10-f	U, N, KU, KN	T, S, R, G	AD, SGD	1e-3,1e-4	1e-3, 1e-4	0, 0.5	1-10
F-MNIST	CNN (s)	Seed	U	T	AD	3e-4	0	0	1-1000
	CNN (s)	Hyp-10-r	U, N, KU, KN	T, S, R, G	AD, SGD	1e-3,1e-4	1e-3, 1e-4	0, 0.5	~ 10
	CNN (s)	Hyp-10-f	U, N, KU, KN	T, S, R, G	AD, SGD	1e-3,1e-4	1e-3, 1e-4	0, 0.5	1-10
SVHN	CNN (s)	Seed	U	T	AD	3e-3	0	0	1-1000
	CNN (s)	Hyp-10-r	U, N, KU, KN	T, S, R, G	AD, SGD	1e-3,1e-4	1e-3, 1e-4, 0	0, 0.3, 0.5	~ 10
	CNN (s)	Hyp-10-f	U, N, KU, KN	T, S, R, G	AD, SGD	1e-3,1e-4	1e-3, 1e-4, 0	0, 0.3, 0.5	1-10
USPS	CNN (s)	Seed	U	T	AD	3e-4	1e-3	0	1-1000
	CNN (s)	Hyp-10-r	U, N, KU, KN	T, S, R, G	AD, SGD	1e-3,1e-4	1e-2, 1e-3	0, 0.5	~ 10
	CNN (s)	Hyp-10-f	U, N, KU, KN	T, S, R, G	AD, SGD	1e-3,1e-4	1e-2, 1e-3	0, 0.5	1-10
CIFAR10	CNN (s)	Seed	KU	G	AD	1e-4	1e-2	0	1-1000
	CNN (s)	Hyp-10-r	U, N, KU, KN	T, S, R, G	AD, SGD	1e-3	1e-2, 1e-3	0, 0.5	~ 10
	CNN (s)	Hyp-10-f	U, N, KU, KN	T, S, R, G	AD, SGD	1e-3	1e-2, 1e-3	0, 0.5	1-10
CIFAR10	CNN (m)	Seed	KU	G	AD	1e-4	1e-2	0	1-1000
	CNN (m)	Hyp-10-r	U, N, KU, KN	T, S, R, G	AD, SGD	1e-3	1e-2, 1e-3	0, 0.5	~ 10
	CNN (m)	Hyp-10-f	U, N, KU, KN	T, S, R, G	AD, SGD	1e-3	1e-2, 1e-3	0, 0.5	1-10
STL (s)	CNN (s)	Seed	KU	T	AD	1e-4	1e-3	0	1-1000
	CNN (s)	Hyp-10-r	U, N, KU, KN	T, S, R, G	AD, SGD	1e-3,1e-4	1e-2, 1e-3	0, 0.5	~ 10
	CNN (s)	Hyp-10-f	U, N, KU, KN	T, S, R, G	AD, SGD	1e-3,1e-4	1e-2, 1e-3	0, 0.5	1-10
STL	CNN (m)	Seed	KU	T	AD	1e-4	1e-3	0	1-1000
	CNN (m)	Hyp-10-r	U, N, KU, KN	T, S, R, G	AD, SGD	1e-3,1e-4	1e-2, 1e-3	0, 0.5	~ 10
	CNN (m)	Hyp-10-f	U, N, KU, KN	T, S, R, G	AD, SGD	1e-3,1e-4	1e-2, 1e-3	0, 0.5	1-10
CIFAR10	RN-18	Seed	KU	R	SGD	1e-4*	5e-4	0	1-1000
CIFAR10	RN-18	Seed	KU	R	SGD	1e-4*	5e-4	0	1-1000
CIFAR10	RN-18	Seed	KU	R	SGD	1e-4*	5e-4	0	1-1000

Prior work discusses the impact of random seeds on properties of model zoos. While [58] use multiple random seeds for the same hyperparameter configuration, [55] explicitly argues against that to prevent information leakage between models from train to test set. To achieve diverse model zoos and disentangle the generating factors (seeds and hyperparameters), we train model zoos in three different configurations, some with random seeds, others with fixed seeds.

`Random Seeds` The first configuration, denoted as `Hyp-10-rand`, varies a broad range of hyperparameters to define a grid of hyperparameters. To include the effect of different random initializations, each of the hyperparameter nodes in the grid is repeated with ten randomly drawn seeds. One model is configured with the combination of hyperparameters and seed, with a total of ten models per hyperparameter node. It is very unlikely for two models in the zoo share the same random seed. With this, we achieve the highest amount of diversity in properties, generating factors and weights.

`Fixed Seeds` The second configuration, denoted as `Hyp-10-fix`, uses the same hyperparameter grid as , but repeats each node with ten fixed seeds $[1, 2, ..., 10]$. Fixing the seeds allows evaluation methods to control for the seed, isolate the influence of hyperparameter choices and still get robust results over 10 repetitions. A side effect of the (desired) isolation of factors of influence is, that fixing the seeds leads to repetitions of the starting point in weight space for models with the same seed and initialization methods. In the beginning of the training, these models may have similar trajectories.

`Fixed Hyperparameters` For the third configuration, denoted as `Seed`, we fix one set of hyperparameters, and repeat that with 1000 different seeds. With that, we achieve zoos that are very diverse in weights and covers a broad range in weight space. These zoos and can be used to evaluate the impact of weights and their starting point on model performance. The hyperparameters for the `Seed` zoos are chosen such that there is still a level of diversity in model performance.

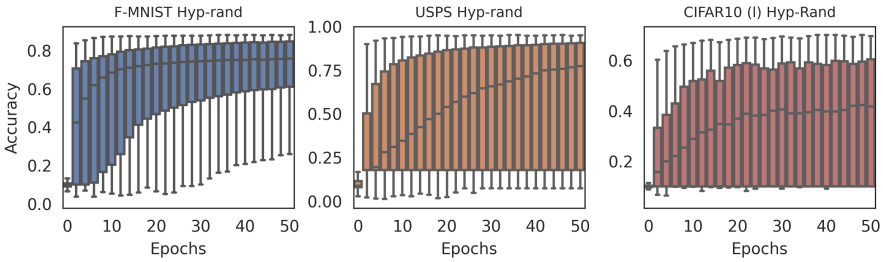

Figure 2: Accuracy distribution over epochs for the `F-MNIST Hyp-rand`, `USPS Hyp-rand` and `CIFAR Hyp-rand` zoos. All zoos show training progress and considerable performance diversity.

3.2 Specification of Generating Factors for Model Zoos

This section describes the systematic specification of the trained model zoos. Multiple generating factors define a configuration $\{\mathcal{D}, \lambda, \mathcal{A}\}$ for the model zoo generation, detailed in Table 1.

Datasets $\mathcal{D}$: We generate model zoos for the following image classification datasets: `MNIST` [30], `Fashion-MNIST` [57], `SVHN` [43], `CIFAR-10` [28], `STL-10` [5], `USPS` [22], `CIFAR-100` [28] and `Tiny Imagenet` [29].

Hyperparameter λ: varied hyperparameters to train models in zoos are: (1) `seed`, (2) `initialization method`, (3) `activation function`, (4) `dropout`, (4) `optimization algorithm`, (5) `learning rate`, and (6) `weight decay`. The batch-size and number of training epoch is kept constant within zoos.

Architecture $\mathcal{A}$: To preserve the comparability within a model zoo, each zoo is generated using a single neural network architecture. One of three standard architectures is used to generate each zoo. Our intention with this dataset is similar to research communities such as Neural Architecture Search (NAS), Meta-Learning or Continual Learning (CL), where initial work started small-scale [64, 47]. Hence, the first two architectures are a small and a slightly larger Convolutional Neural Network (CNN), both have three convolutional and two fully-connected layers, but different numbers of channels (details in Appendix A). The third architecture is a standard ResNet-18 [20]. The (1) `small CNN` has a total of $2'464$-$2'864$ parameters, the (2) `medium CNN` has $10'853$ parameters, the (3) ResNet-18 has 11.2M-11.3M parameters.

Compared to (1), the medium architecture (2) provides additional diversity to the collection of model zoos and performs significantly better on more complex datasets `CIFAR-10` and `STL-10`. These architectures are similar to the one used in [51]. The ResNet-18 architecture is included to apply the model zoo blueprint to models of the widely used ResNet family and so facilitate research on populations of real-world sized models.

3.3 Training of Model Zoos

Neural network models are trained from the previously defined three configurations $\{\mathcal{D}, \lambda, \mathcal{A}\}$ (Seed, Hyp-10-rand, Hyp-10-fix, see Sec 2.1). With the *8 image datasets* and the three configurations, this results in *27 model zoos*. The zoos include a total of around *50'360* unique neural network models.

Training Protocol: Every model in the collection of zoos is trained according to the same protocol. We keep the same train, validation and test splits for each zoo, and train each model for 50 epochs with gradient descent methods (SGD+momentum or ADAM). At every epoch, the model checkpoint as well as accuracy and loss of all splits are recorded. Validation and test performance are also recorded before the first training epoch. This makes 51 checkpoints per model training trajectory including the starting checkpoint representing the model initialization before training starts. The ResNet-18 zoos on CIFAR100 and Tiny Imagenet require more updates and are trained for 60 epochs. In total, this results in a set of *2'585'360* collected model states.

Splits: To enable comparability, this set of models is split into `training` (70%), `validation` (15%), and `test` (15%) subsets. This split is done such that all individual checkpoints of one model training (i.e., the 51 checkpoints per training) is entirely in either `training`, `validation`, or `test` and therefore no information is leaked between these subsets.

Table 2: Analysis of the diversity of our 27 model zoos (one row per zoo). Mean (std) values in % per zoo, computed on the last epoch. Agreement is computed using samples from the test split of the image dataset pairwise over the entire zoo. Higher agreement values indicate more uniform behavior and less behavioral diversity. Distance in weight space are computed pairwise over the entire zoo. Higher distance values indicate larger diversity in weight space.

			Performance	Agreement		Weights		
Dataset	Architecture	Config	Accuracy	κ_{aggr}	κ_{cka}	$\mathbf{w}$	l2-dist	cos dist
MNIST	CNN (s)	Seed	91.1 (0.9)	88.5 (1.3)	77.2 (5.2)	18.9 (58.4)	124.1 (4.9)	77.1 (4.1)
	CNN (s)	Hyp-10-r	79.9 (30.7)	67.7 (35.5)	58.6 (25.9)	0.4 (46.5)	150.6 (66.5)	98.8 (7.2)
	CNN (s)	Hyp-10-f	80.3 (30.3)	68.3 (35.3)	58.8 (25.7)	0.3 (46.7)	149.7 (66.8)	97.7 (10.0)
F-MNIST	CNN (s)	Seed	72.7 (1.0)	79.8 (2.6)	82.3 (12.6)	22.6 (55.6)	122.0 (4.9)	74.5 (4.4)
	CNN (s)	Hyp-10-r	68.4 (23.7)	59.9 (29.1)	64.6 (23.5)	1.0 (46.0)	149.6 (62.2)	99.2 (6.8)
	CNN (s)	Hyp-10-f	68.7 (23.4)	60.4 (28.7)	64.6 (22.7)	0.9 (46.3)	148.5 (61.9)	97.9 (9.9)
SVHN	CNN (s)	Seed	71.1 (8.0)	67.2 (10.3)	67.7 (15.7)	7.1 (113.7)	137.6 (8.3)	94.5 (5.1)
	CNN (s)	Hyp-10-r	35.9 (24.3)	61.6 (35.9)	17.8 (28.0)	1.4 (42.2)	170.5 (149.4)	83.6 (30.4)
	CNN (s)	Hyp-10-f	36.0 (24.4)	61.4 (36.0)	18.1 (27.9)	1.3 (42.2)	170.0 (149.0)	83.2 (30.7)
USPS	CNN (s)	Seed	87.0 (1.7)	87.3 (2.2)	86.7 (6.3)	8.2 (26.9)	123.1 (5.2)	75.9 (5.0)
	CNN (s)	Hyp-10-r	64.7 (30.8)	55.3 (31.4)	50.9 (30.5)	2.1 (39.6)	155.5 (92.6)	99.1 (8.9)
	CNN (s)	Hyp-10-f	65.0 (30.7)	55.4 (31.3)	50.4 (30.4)	1.9 (40.1)	154.2 (93.1)	97.3 (13.7)
CIFAR10	CNN (s)	Seed	48.7 (1.4)	65.7 (3.1)	72.9 (11.3)	1.1 (11.0)	138.7 (5.6)	96.3 (5.1)
	CNN (s)	Hyp-10-r	35.1 (16.3)	33.3 (22.9)	47.5 (34.0)	-0.2 (17.0)	155.6 (71.0)	97.5 (10.8)
	CNN (s)	Hyp-10-f	35.1 (16.2)	33.3 (22.8)	47.3 (34.2)	-0.2 (16.9)	155.3 (70.0)	97.2 (11.1)
CIFAR10	CNN (m)	Seed	61.5 (0.7)	76.0 (1.6)	92.4 (1.7)	0.1 (18.2)	137.0 (7.9)	94.1 (9.2)
	CNN (m)	Hyp-10-r	39.6 (21.8)	34.5 (27.1)	43.2 (36.5)	-0.4 (23.0)	158.9 (79.9)	98.6 (12.2)
	CNN (m)	Hyp-10-f	39.6 (21.7)	34.4 (26.7)	42.8 (37.8)	-0.4 (22.9)	158.1 (77.2)	98.0 (13.1)
STL	CNN (s)	Seed	39.0 (1.0)	48.4 (3.0)	81.5 (3.9)	-0.1 (19.1)	141.2 (5.0)	99.8 (4.2)
	CNN (s)	Hyp-10-r	23.1 (12.3)	23.4 (20.9)	39.0 (30.7)	3.0 (40.0)	158.7 (107.3)	98.7 (10.9)
	CNN (s)	Hyp-10-f	23.0 (12.2)	23.3 (21.1)	38.1 (30.0)	3.0 (39.8)	157.1 (107.2)	96.8 (16.3)
STL	CNN (m)	Seed	47.4 (0.9)	53.9 (2.2)	83.3 (2.3)	0.1 (26.6)	141.3 (6.0)	99.9 (5.8)
	CNN (m)	Hyp-10-r	24.3 (14.7)	23.2 (24.2)	34.1 (30.0)	2.3 (45.7)	159.3 (103.0)	99.1 (12.5)
	CNN (m)	Hyp-10-f	24.4 (14.7)	23.7 (24.5)	34.6 (30.3)	2.3 (46.5)	157.4 (104.1)	97.6 (20.1)
CIFAR10	ResNet-18	Seed	92.1 (0.2)	93.4 (0.7)	–.- (-.-)	-0.01 (1.7)	122.1 (3.9)	72.2 (2.3)
CIFAR100	ResNet-18	Seed	74.2 (0.3)	77.6 (1.2)	–.- (-.-)	-0.1 (1.6)	130.8 (4.1)	83.1 (2.6)
Tiny ImageNet	ResNet-18	Seed	63.9 (0.7)	66.1 (1.9)	–.- (-.-)	-0.1 (1.9)	125.4 (4.9)	77.1 (3.0)

Sparsified Model Zoo Twins: Model sparsification is an effective method to reduce computational cost of models. However, methods to sparsify models to a high degree while preserving the performance are still actively researched [21]. In order to allow systematic studies of sparsification, we are extending the model zoos with sparsified *model zoo twins* serving as counterparts to existing zoos in the dataset. Using Variational Dropout (VD) [41], we sparsify the trained models from existing model zoos. VD generates a sparsification trajectory for each model, along which we track the performance, degree of sparsity and the sparsified checkpoint. With 25 sparsification epochs, this yields 1'259'000 sparsification model states.

3.4 Data Management and Accessibility of Model Zoos

The model zoos are made publicly available in an accessible, standardized, and well documented way to the research community under the Creative Commons Attribution 4.0 license (CC-BY 4.0). We ensure the technical accessibility of the data by hosting it on Zenodo, where the data will be hosted for at least 20 years. Further, we take steps to reduce access barriers by providing code for data loading and preprocessing, to reduce the friction associated with analyzing of the raw zoo files. All code can be found on the model zoo website www.modelzoos.cc. To ensure conceptional accessibility, we include detailed insights, visualizations and the analysis of the model zoo (Sec. 4) with each zoo. Further details can be found in Appendix B.